

1. Determine whether the following are true or false. Explain your answer or give a counterexample.

a. $\text{proj}_{\mathbf{v}} \mathbf{u} = \text{proj}_{\mathbf{u}} \mathbf{v}$.

- b. If nonzero vectors $\mathbf{v}$ and $\mathbf{u}$ have the same magnitude they make equal angles with $\mathbf{u} + \mathbf{v}$.

- c. If $\mathbf{u}$ is orthogonal to $\mathbf{v}$ and $\mathbf{v}$ is orthogonal to $\mathbf{w}$, then $\mathbf{u}$ is orthogonal to $\mathbf{w}$.

- d. The vectors orthogonal to $\langle 1, 1, 1 \rangle$ lie on the same line.

- e. If $\text{proj}_{\mathbf{v}} \mathbf{u} = \mathbf{0}$, then vectors $\mathbf{u}$ and $\mathbf{v}$ (both nonzero) are orthogonal.

2. Find all vectors orthogonal to $\mathbf{v}$.

f. $\mathbf{v} = \langle 3, 4, 0 \rangle$

g. $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$